Question:

Assume the label **Start** is located at address **70000**.

MIPS Instruction with the instruction addresses are:

Address	Instruction	Format
	Start:	
70000	addi $t3, $zero, 15	I - Format
70004	sub $t4, $t4, $t3	R-Format
70008	sw $t4, 0($s2)	I - Format
70012	bne $t4, $zero, Done	I - Format
70016	add $s2, $s2, $t1	R-Format
70020	j Start	J-Format
70024	Done: ...	

a) Identify the format of each instruction (R-format, I-format, or J-format).

b) Calculate the branch offset stored in the bne instruction.

c) Determine the target address encoded in the j Start instruction.

b) $\text{offset} = \frac{(\text{Target} - (\text{PC} + 4))}{4}$

$= 70024 - \left(\frac{70012}{4}\right) = 2$

c) start = 70000

$\frac{\text{Target address}}{4} = \frac{70000}{4} = 17500$

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